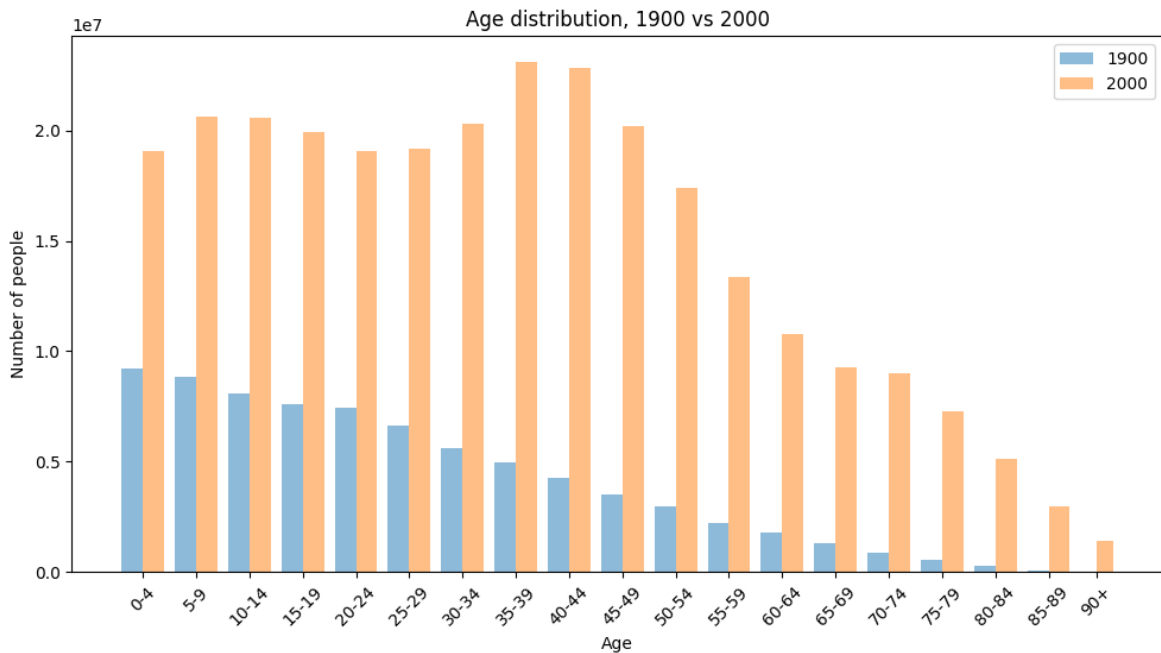


# Homework 1

## Problem 1

My question: How does population compare between years 1900 and 2000, by age-range?



Population distribution by age for the years 1900 and 2000

Figure 1: census graph

From the graph, it looks like compared to the year 2000, there were a lot less people in 1900 in general. Also, the population distribution in 1900 indicates that most people were in the younger age bucket, as the graph is right-skewed. Although the population distribution in the year 2000 is also right-skewed, it looks like it has a bimodal distribution, with peaks in the 5-14 age range and 35-44 age range. Also compared to the year 1900, it looks like there are a

lot more people in general, especially in the later age ranges (60+). In my code below, I first set up the given bins and then split the CSV file into years 1900 and 2000, where I isolated the “Age” and “People” columns (disregard gender). Finally, I plotted the histogram with side-by-side bars of the population in year 2000 vs. year 1900 for each age bucket.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

census = pd.read_csv("census1900-2000.csv")

# Create histogram with age buckets of 0-4, 5-9, 10-14, ..., 90+
age_bins = [0, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50,
            55, 60, 65, 70, 75, 80, 85, 90, 94]
age_labels = ['0-4', '5-9', '10-14', '15-19', '20-24',
              '25-29', '30-34', '35-39', '40-44', '45-49',
              '50-54', '55-59', '60-64', '65-69', '70-74',
              '75-79', '80-84', '85-89', '90+']

# Further split into either 1900 or 2000
census_1900 = census.loc[census['Year'] == 1900, ["Age", "People"]]
census_2000 = census.loc[census['Year'] == 2000, ["Age", "People"]]

# Plot w/ side-by-side bars for 1900 and 2000
fig, ax = plt.subplots(figsize=(10, 6))
ax.hist([census_1900["Age"], census_2000["Age"]],
        bins=age_bins,
        weights=[census_1900["People"], census_2000["People"]],
        label=["1900", "2000"],
        alpha=0.5)
bin_centers = [(age_bins[i] + age_bins[i + 1]) / 2 for i in range(len(age_bins) - 1)]
ax.set_xticks(bin_centers)
ax.set_xticklabels(age_labels, rotation=45)

ax.set_xlabel("Age")
ax.set_ylabel("Number of people")
ax.set_title("Age distribution, 1900 vs 2000")
ax.legend()
fig.text(
    0.5, 0.01,
    'Population distribution by age for the years 1900 and 2000',
    ha='center', fontsize=10, style='italic'
)
```

```
fig.tight_layout(rect=[0, 0.05, 1, 1])

# Save plot as .png file in folder
fig.savefig("age_distribution_1900_vs_2000.png")
```

## Problem 2

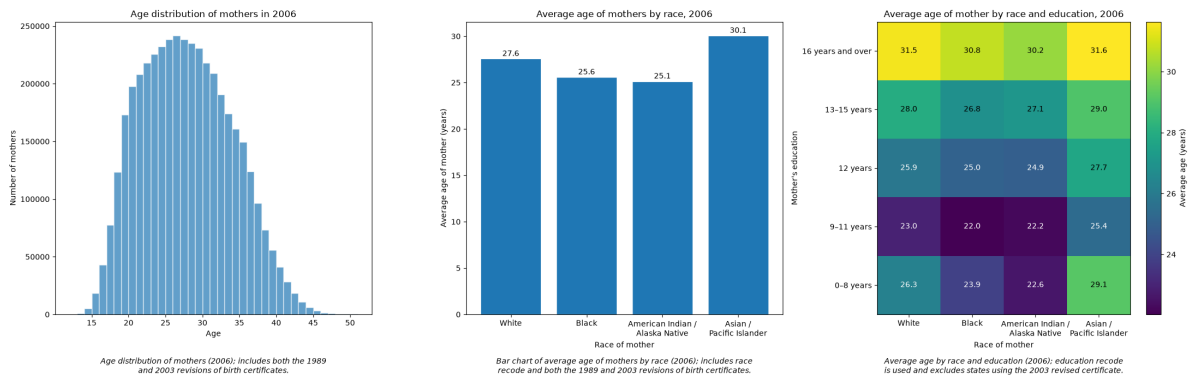


Figure 2: mother graph

```
# Age distribution of mothers in 2006 (graph 1)
axs[0].hist(df["MAGER"], bins=np.arange(12, 52), alpha=0.7, edgecolor="white")

# Average age of mothers by race, 2006 (graph 2)
axs[1].bar(avg_age.index, avg_age.values, edgecolor="white")

# Average age by race and education, 2006 (graph 3)
fig.colorbar(im, ax=axs[2], label="Average age (years)")
```

## AI Statement

I used Claude (Anthropic) while preparing this homework. Specifically:

- Data setup: Claude extracted the field positions from the 2006 NCHS Natality User Guide and generated the KEY2006.txt key file, plus a helper script (loadnat2006.py) that uses this key to read the fixed-width Nat2006us.dat file into a pandas dataframe.
- Variable guidance: Claude explained which variables are available for all states versus only states using the revised or unrevised birth certificate, and how to handle missing-value codes (e.g., 9999 for birth weight).

- Debugging: I wrote the initial plotting code, and Claude helped fix errors, including a `pd.NA` type error, indexing a single matplotlib Axes, and overlapping captions.
- Visualization: Claude suggested and wrote code for creating the bar chart of average age by race, and replacing a race/education scatter plot with a heatmap which worked better for understanding the relationship between the three variables.

I chose the questions to investigate and the variables to plot, ran all code on the data myself, and checked the results and variable definitions against the User Guide. The paragraph interpretation of the graph in problem 1 is my own.